Interview Questions and Answers
AI, Machine Learning & Deep Learning

1. AI Fundamentals

Q1. What is Artificial Intelligence (AI)?

A1. AI is the field of creating systems that can perform tasks requiring human-like intelligence such as perception, reasoning, learning, and decision-making.

Q2. AI vs Machine Learning vs Deep Learning?

A2. AI is the broad field. ML is a subset where systems learn from data. DL is a subset of ML that uses multi-layer neural networks.

Q3. What are the types of AI?

A3. Narrow AI (task-specific), General AI (human-level across tasks—still not achieved), and Super AI (beyond human—hypothetical).

Q4. What is an intelligent agent?

A4. An entity that perceives its environment and takes actions to maximize a goal (reward/utility).

Q5. What is supervised learning?

A5. Learning a mapping from inputs to labeled outputs using training examples.

Q6. What is unsupervised learning?

A6. Finding patterns/structure in unlabeled data (e.g., clustering, dimensionality reduction).

Q7. What is reinforcement learning?

A7. An agent learns by interacting with an environment to maximize cumulative reward.

Q8. What is overfitting?

A8. When a model learns noise in training data and performs poorly on unseen data.

Q9. What is underfitting?

A9. When a model is too simple to capture patterns and performs poorly on both train and test.

Q10. What is the bias-variance tradeoff?

A10. Bias is error from oversimplification; variance is error from sensitivity to training data. The goal is a balance for best generalization.

2. Data & ML Workflow

Q1. Typical ML pipeline steps?

A1. Problem definition → data collection → cleaning/preprocessing → feature engineering → model training → evaluation → tuning → deployment → monitoring.

Q2. What is train/validation/test split?

A2. Train for learning parameters, validation for tuning/hyperparameters, and test for final unbiased evaluation.

Q3. What is cross-validation?

A3. A technique (e.g., k-fold) to evaluate a model reliably by training/testing on multiple splits.

Q4. What is feature engineering?

A4. Creating/transforming input variables to improve model performance (e.g., scaling, encoding, interaction terms).

Q5. Why is scaling important?

A5. Algorithms like KNN, SVM, and gradient descent-based models are sensitive to feature magnitude.

Q6. One-hot encoding vs label encoding?

A6. One-hot creates binary columns for categories. Label encoding assigns integers; it may imply order (use carefully).

Q7. How do you handle missing values?

A7. Remove rows/columns, impute with mean/median/mode, model-based imputation, or use algorithms that handle missing values.

Q8. How do you handle imbalanced datasets?

A8. Use resampling (SMOTE/undersampling), class weights, threshold tuning, and appropriate metrics (F1, PR-AUC).

Q9. What is data leakage?

A9. Using information from outside the training data (or from the future) that inflates performance; avoid by proper splitting and pipeline usage.

Q10. What are hyperparameters?

A10. Model settings set before training (e.g., learning rate, depth, C in SVM), tuned using validation.

3. Supervised Learning Algorithms

Q1. What is linear regression used for?

A1. Predicting continuous values; it fits a line/plane minimizing error (often MSE).

Q2. What is logistic regression used for?

A2. Binary/multiclass classification; outputs probabilities using sigmoid/softmax.

Q3. KNN: how does it work?

A3. Classifies/predicts based on majority/average of k nearest data points using a distance metric.

Q4. Decision tree pros/cons?

A4. Pros: interpretable, handles non-linear. Cons: can overfit; unstable with small data changes.

Q5. Random Forest vs Decision Tree?

A5. Random Forest is an ensemble of trees trained on bootstrapped samples + feature randomness, reducing overfitting and improving generalization.

Q6. What is boosting (e.g., XGBoost)?

A6. Ensemble method where models are trained sequentially to correct previous errors; strong performance on tabular data.

Q7. SVM intuition?

A7. Finds a separating hyperplane with maximum margin; kernels handle non-linear separation.

Q8. Naive Bayes assumption?

A8. Features are conditionally independent given the class—often works well for text classification.

Q9. What is regularization?

A9. Penalizes complex models to reduce overfitting (L1/L2).

Q10. L1 vs L2 regularization?

A10. L1 (Lasso) can produce sparse weights (feature selection). L2 (Ridge) shrinks weights smoothly.

4. Unsupervised Learning

Q1. What is clustering?

A1. Grouping similar data points without labels.

Q2. K-means: key idea?

A2. Partitions data into k clusters by minimizing within-cluster squared distance to centroids.

Q3. Limitations of K-means?

A3. Needs k, sensitive to outliers and initialization, assumes spherical clusters.

Q4. Hierarchical clustering?

A4. Builds clusters as a tree (dendrogram) using agglomerative or divisive approaches.

Q5. DBSCAN advantage?

A5. Finds arbitrary-shaped clusters and detects outliers; does not require k.

Q6. What is PCA?

A6. Dimensionality reduction technique that projects data onto directions of maximum variance.

Q7. PCA vs t-SNE/UMAP?

A7. PCA is linear and good for compression; t-SNE/UMAP are non-linear and better for visualization of complex structures.

Q8. What is anomaly detection?

A8. Identifying rare/abnormal patterns; methods include Isolation Forest, One-Class SVM, statistical thresholds.

Q9. What is association rule mining?

A9. Finding relationships like {bread} → {butter} using support, confidence, and lift.

Q10. What is feature scaling impact on clustering?

A10. Distance-based clustering (K-means) is sensitive to scale, so scaling is important.

5. Model Evaluation Metrics

Q1. Define accuracy.

A1. Accuracy = (TP + TN) / Total. Good when classes are balanced.

Q2. Precision vs Recall?

A2. Precision = TP/(TP+FP). Recall = TP/(TP+FN).

Q3. What is F1-score?

A3. Harmonic mean of precision and recall: F1 = 2PR/(P+R).

Q4. What is confusion matrix?

A4. A table of TP, FP, TN, FN for classification performance analysis.

Q5. ROC-AUC meaning?

A5. Measures tradeoff between TPR and FPR across thresholds; good for ranking performance.

Q6. PR-AUC vs ROC-AUC?

A6. PR-AUC is more informative for imbalanced datasets where positives are rare.

Q7. Regression metrics?

A7. MAE, MSE, RMSE, and R^2 are common regression evaluation metrics.

Q8. What is R^2?

A8. Proportion of variance explained by the model; 1 is perfect, 0 equals predicting mean.

Q9. What is log loss?

A9. Penalizes wrong confident probabilities in classification.

Q10. Why separate validation and test sets?

A10. To avoid over-tuning; test set provides unbiased final estimate.

6. Deep Learning Basics

Q1. What is a neural network?

A1. A model composed of layers of neurons that learn representations using weighted sums and non-linear activations.

Q2. What is an activation function?

A2. Introduces non-linearity; common ones are ReLU, sigmoid, tanh, softmax.

Q3. Why is ReLU popular?

A3. Simple, reduces vanishing gradients, speeds training in deep networks.

Q4. What is backpropagation?

A4. Algorithm to compute gradients of loss w.r.t. weights using chain rule.

Q5. What is gradient descent?

A5. Optimization algorithm that updates weights opposite to gradient to minimize loss.

Q6. Batch vs mini-batch vs stochastic GD?

A6. Batch uses all data, stochastic uses one sample, mini-batch uses a small subset (most common).

Q7. What is learning rate?

A7. Step size for weight updates; too high diverges, too low trains slowly.

Q8. What is dropout?

A8. Regularization technique that randomly drops neurons during training to reduce overfitting.

Q9. What is batch normalization?

A9. Normalizes layer inputs to stabilize and speed training; often improves performance.

Q10. What are common DL optimizers?

A10. SGD with momentum, Adam, RMSProp. Adam is widely used for fast convergence.

7. CNN, RNN, Transformers

Q1. What is a CNN used for?

A1. Primarily for images and spatial data; uses convolution filters to learn local patterns.

Q2. What is convolution and kernel?

A2. A kernel (filter) slides over input to produce feature maps capturing patterns like edges.

Q3. What is pooling?

A3. Downsampling operation (max/avg) to reduce spatial size and computation and add invariance.

Q4. What is an RNN used for?

A4. Sequence data (text/time series) by maintaining hidden state over time.

Q5. Why LSTM/GRU over vanilla RNN?

A5. They reduce vanishing gradient issues and learn longer dependencies.

Q6. What is attention?

A6. Mechanism that lets the model focus on relevant parts of the input when generating outputs.

Q7. What is a Transformer?

A7. A model architecture based on self-attention (no recurrence), used for NLP and many tasks.

Q8. What are embeddings?

A8. Dense vector representations of tokens/items capturing semantic similarity.

Q9. What is positional encoding?

A9. Adds information about token order to transformers since attention alone is order-agnostic.

Q10. When to use CNN vs RNN vs Transformer?

A10. CNN: images/local patterns. RNN/LSTM: sequences with temporal order. Transformer: long-range dependencies, modern NLP and multimodal.

8. Generative AI & LLM Basics (Bonus for Freshers)

Q1. What is Generative AI?

A1. Models that can generate new content (text, images, code) similar to training data.

Q2. What is an LLM?

A2. Large Language Model trained on massive text to predict next token; used for chat, summarization, coding, etc.

Q3. What is prompt engineering?

A3. Designing prompts to get reliable outputs (role, context, constraints, examples).

Q4. What is hallucination in LLMs?

A4. When a model generates plausible but incorrect information; mitigate using retrieval, citations, and verification.

Q5. What is RAG?

A5. Retrieval-Augmented Generation: retrieve relevant documents then generate grounded answers.

Q6. What are tokens?

A6. Small units of text used by language models; inputs/outputs are tokenized.

Q7. What is temperature?

A7. Controls randomness: low = more deterministic, high = more creative.

Q8. What is fine-tuning?

A8. Training a pre-trained model further on task-specific data to improve behavior.

Q9. What is embedding similarity used for?

A9. Semantic search, clustering, recommendations, and retrieval in RAG systems.

Q10. How do you evaluate a GenAI app?

A10. Use correctness checks, human evaluation, grounding tests, latency/cost monitoring, and safety checks.

9. Practical & Project Interview Questions

Q1. How do you explain an ML project in interviews?

A1. State problem → dataset → preprocessing → model selection → metrics → results → deployment/next steps.

Q2. How did you choose the algorithm?

A2. Based on data type, baseline results, interpretability needs, and validation performance.

Q3. How did you handle overfitting?

A3. Used regularization, simpler model, more data, dropout/early stopping, and cross-validation.

Q4. What is early stopping?

A4. Stop training when validation performance stops improving to prevent overfitting.

Q5. How do you tune hyperparameters?

A5. Grid search, random search, Bayesian optimization using validation or cross-validation.

Q6. How do you deploy an ML model?

A6. Package model + preprocessing pipeline, expose API (Flask/FastAPI), containerize (Docker), monitor in production.

Q7. How do you monitor a model in production?

A7. Track data drift, prediction drift, latency, errors, and periodic re-training needs.

Q8. What is data drift vs concept drift?

A8. Data drift: input distribution changes. Concept drift: relationship between input and target changes.

Q9. How do you ensure reproducibility?

A9. Fix random seeds, version data/code, log experiments, use pipelines and environment management.

Q10. What is a baseline model and why important?

A10. A simple reference model to compare improvements; prevents overengineering.